

# [T]-Theory: A Universal Theory of Everything

One Equation. 61 Decades. Mind, Body, and Cosmos.

*Why:* complex systems become intelligible when their many local interactions are read as one evolving field. [T]-Theory treats emergence, criticality, and long-range coordination as different expressions of a shared propagator and attractor landscape.

## Proved (Lean 4, 0 sorries):

- SHO = Green's function (derived, not postulated as a primitive)
- 11D manifold = minimum geometry for a conscious vertebrate organism
- Scale invariance  $10^{-35}\text{m} \rightarrow 10^{26}\text{m}$  (61 orders of magnitude)
- Theory is a proposed fixed-point attractor of its own subject matter (see the [T]-Theory phenomena paper)

## Numerical hits (Planck 2018):

- $\Omega_\Lambda = 7/11$  (93.2%) — dark energy as compactified vacuum
- $\Omega_c = 3/11$  (97.1%) — dark matter as spatial vacuum; gravity only; no direct detection possible
- $w = -1$  exact — live knife-edge test: DESI DR1 & Euclid

[T]-Theory accounts for 95.5% of the total mass-energy of the universe via a zero-parameter geometric derivation.

## I · The Master Field Equation

The Universal Somatic Field is governed by a single structural equation — the Green's function of a tensor field derived from M-theory compactification.

### Stationary form (Helmholtz):

$$(\nabla^2 + k^2) G(x, x') = -\delta^3(x - x')$$

### Time-dependent form (d'Alembertian):

$$\left( \frac{1}{v_s^2} \partial_t^2 - \nabla^2 + k^2 \right) \Phi_{\mu\nu}(x, t) = -J_{\mu\nu}(x, t)$$

**Retarded propagator** — causal,  $\tau = t - t' > 0$ :

$$G_R(r, \tau) = \frac{v_s}{4\pi r} e^{-kv_s\tau} \delta(\tau - r/v_s) \theta(\tau)$$

### Somatic Memory Kernel:

$$K(\tau) = K_0 e^{-\tau/\tau_m} \theta(\tau), \quad \tau_m = \frac{1}{kv_s}$$

**General solution** (field = integral of all past sources):

$$\Phi(x, t) = \Phi^{(0)}(x, t) + \int_{-\infty}^t dt' \int d^3x' G_R J$$

The **effective mass**  $k$  sets the spatial correlation length  $\ell = 1/k$ . As  $k \rightarrow 0$  (near consciousness threshold  $T_c$ ),  $\ell \rightarrow \infty$ : global integration onset.

The **field temperature**  $T_{\text{field}}$  controls how freely the field explores the attractor landscape. Therapeutic window:  $T_{\text{field}} \in [T_{\text{min}}, T_{\text{max}}]$ .

## II · Type-Theoretic Architecture (HoTT / Lean 4)

The soma-field has a rigorous type-theoretic formulation in Homotopy Type Theory, machine-verified in Lean 4 using Mathlib and Physlib.

**The  $\Sigma$ -type** — soma-field as a fiber bundle over the 20-scale base:

$$\text{SomaField} \equiv \sum_{(\sigma : \text{Scale}_{20})} \text{Substrate}(\sigma)$$

where  $\text{Substrate}(\sigma) : \text{Type}$  is the physical substrate at scale  $\sigma$ . The base space is the 20-point scale hierarchy; each fiber is a field configuration.

**The Zoom Operator** — dependent type constructor between adjacent fibers:

$$\Lambda : (\sigma : \text{Scale}_{20}) \rightarrow \text{Substrate}(\sigma) \rightarrow \text{Substrate}(\sigma + 1)$$

**Causality as a proof argument** — the retarded propagator's full type:

$$G_R : (t \ t' : \text{Time}) \rightarrow (t' < t) \rightarrow \text{Space} \rightarrow \text{Space} \rightarrow \text{Field}$$

The inequality  $t' < t$  is a *proof term* — the Lean 4 kernel enforces the arrow of time at compile time. A causal violation is a type error, not a runtime error.

**The full USF type:**

$$\text{USF} \equiv \sum_{(\sigma : \text{Scale}_{20})} (\text{Substrate}(\sigma) \times G_R(\sigma))$$

**BFSS cortex** — coordinates emerge from Hermitian